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ALDWIN C. SUMALINOG

ExamInsight: ATTAINMENT & PROGRESS TRACKING IN YEAR 2 INTERNAL ASSESSMENTS AND EXTERNAL BENCHMARK TESTS AT PRISTINE PRIVATE SCHOOL

Thesis/Dissertation Adviser:

REINALD ADRIAN PUGOY
Faculty of Information and Communication Studies

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ExamInsight: ATTAINMENT & PROGRESS TRACKING IN YEAR 2 INTERNAL ASSESSMENTS AND EXTERNAL BENCHMARK TESTS AT PRISTINE PRIVATE SCHOOL

is hereby accepted by the Faculty of Information and Communication Studies in partial fulfillment of the requirements for the Master of Information Systems.

REINALD ADRIAN PUGOY
Adviser

_____
Date

BLANCAFLOR P. ARADA
Program Chair

_____
Date

DIEGO S. MARANAN, Ph.D.
Dean
Faculty of Information and
Communication Studies

_____
Date

ABSTRACT

ExamInsight, a proposed system, aims to streamline the tracking of attainment and progress in Year 2 internal assessments and external benchmark tests at Pristine Private School through an automated solution leveraging AI technology.

The rationale behind this initiative stems from the limited capacity of providing comprehensive analysis and interpretation of the current conventional method, relying heavily on spreadsheets and formula-based calculations. While functional, these methods are time-consuming and fail to provide insightful analytics on student performance. The proposed system aims to address these limitations by automating tracking processes and utilizing AI technology, like ChatGPT for advanced data analysis and interpretation. Key features considered for the system include:

- Individual student performance tracking over time
- Aggregated class/grade-level performance data
- Ability to disaggregate data by groups/subgroups (ex. gender, year group)
- Comprehensive reporting and visualization tools

Development of this system will make use of the agile project management technique, such as personal SCRUM through a phased approach, ensuring a systematic and thorough execution of the project, from conceptualization to deployment. The process begins with a 4-week planning and analysis phase, laying the groundwork for the project. The project will move into a 20-week design and development phase, focusing on backend, front end, and API integration. A comprehensive 5-week testing phase will be conducted prior to a 3-week deployment phase.

Leveraging AI, the system can generate comprehensive reports and visual representations, offering valuable insights into student performance [1]. Adopting this system will overcome limitations of the current conventional method and ensure data privacy.

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Chapter I

THE PROBLEM DOMAIN

A. Statement of the Problem

Traditional method of tracking student attainment and progress rely heavily on manual data entry, spreadsheets, and formula-based calculations. This conventional approach is time-consuming, error-prone, and lacks advanced data analysis and interpretation capabilities. Tracking attainment and progress in Year 2 at Pristine Private School uses such conventional method where it is proven to have faced several limitations:

- ✓ Inefficiency – manual data entry consumes valuable time
- ✓ Limited insights – provide basic performance metrics only but lack deeper insights into students' performance
- ✓ Data discrepancies – discrepancies between internal assessments and external benchmark tests may exist and may go unnoticed

With this, there is a pressing need for an automated, AI-driven solution that can centralize data, streamline tracking processes, and provide insightful analytics.

B. Background and Objectives of the Project

The project was initiated due to the significant gaps in the current approach to tracking attainment and progress. Relying on the conventional method with lack of in-depth analysis on assessment data makes it difficult for teachers and school administrators to efficiently identify areas where students need improvement and to implement timely interventions.

Objectives

To address the challenges and limitations of the current approach, the proposed system aims to achieve the following objectives:

- ✓ *Automate Data Tracking* – streamline the process and reduce reliance on manual data entry and spreadsheets

- ✓ *Provide comprehensive analytics* – harness AI technology to offer detailed and insightful analysis on student performance
- ✓ *Ensure data consistency* – incorporate validation checks to maintain accuracy across assessment data

C. Significance and Scope of the Project

The proposed system holds significant importance in addressing the shortcomings of the current approach to tracking attainment and progress at Pristine Private School. It will encompass several key functionalities designed to address the limitations of the current conventional method. The scope of the project includes:

Data Automation

- ✓ **Automate data tracking**. The system will include functionalities to streamline the collection and management of assessment data from both internal and external sources, minimizing manual data entry and reducing errors.
- ✓ **Data import facility**. The system will be developed to include a feature that will enable the importation of assessment data from standard formats like CSV and Excel, ensuring easy integration with existing data sources.

Comprehensive Analytics

- ✓ **AI-driven analysis**. The system will integrate AI technology like ChatGPT to provide detail and insightful analysis of student performance, highlighting areas to improve and overall academic progress.
- ✓ **Interactive dashboards**. The proposed system will include user-friendly dashboards that will present analytics and reports in a visually appealing and easily interpretable manner for teachers and school administrators.

Data Consistency and Validation

- ✓ **Validation checks and error handling**. The system will implement validation mechanism to ensure accuracy of assessment data and error detection and correction functionalities to maintain data integrity.

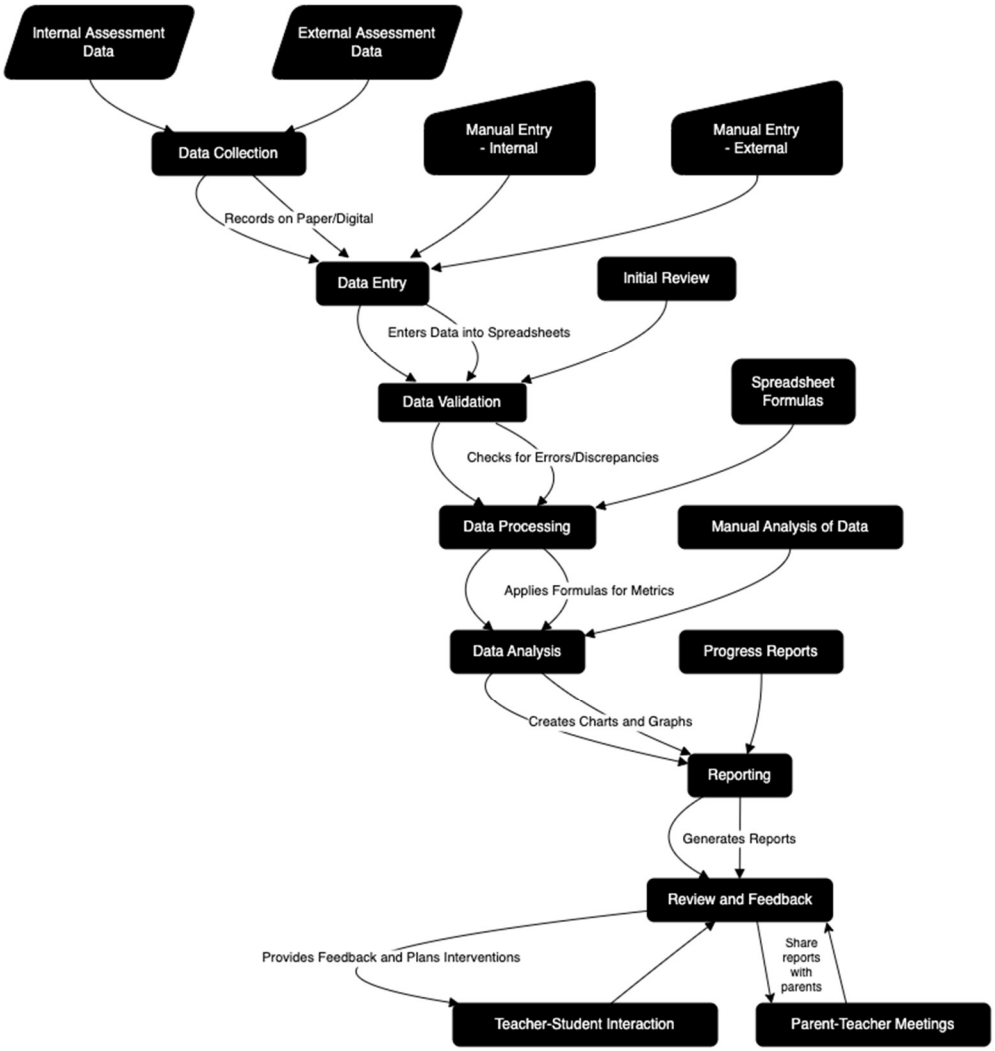
User Management and Access Control

- ✓ **Account and access control**. The system will provide administrative functionalities for managing user accounts, including adding new users, modifying user roles and privileges, and ensuring secure access to system features and data.
- ✓ **Secure authentication**. The proposed system will implement secure login functionalities to protect user data and system access.

D. Documentation of Existence and Seriousness of the Problem

The existing system for tracking student attainment and progress involves multiple steps, which are largely manual and dependent on spreadsheets and formula-based calculations.

Figure 1. Existing Process Flow of Tracking Student Attainment & Progress



Process Flow Details

Data Collection

- *Internal Assessments*. Teachers administer end of term internal assessments and scores are recorded on paper or spreadsheets
- *External Benchmark* Tests. NGRT tests are conducted, and results are provided to the school in digital format (CSV, Excel)

Data Entry

- *Manual Data Entry*. Teachers manually input scores from internal assessments into spreadsheets containing formulas for required calculations. External benchmark tests results are transferred from the provided digital format (CSV, Excel) into another spreadsheet to collate with other assessment data.

Data Validation

- *Initial Review*. Office staff review the manually entered data to check for errors or discrepancies. Corrections are made as needed to ensure data accuracy.

Data Processing

- *Spreadsheet Formulas*. Formulas are created and entered to calculate averages, percentages, and other metrics. Custom formulas are applied to generate specific reports and insights.

Data Analysis

- *Manual Analysis*. Teachers and school administrators manually analyze the data to identify trends and performance patterns. Basic visualizations, such as charts and graphs are created using spreadsheet tools.

Reporting

- *Progress Reports*. Progress reports for individual students are generated by extracting data from spreadsheets. Some reports are printed or shared digitally with parents. Summary reports for classes and year group are generated and compiled for internal review, staff meetings and for strategic planning.

Review and Feedback

- *Teacher-Student Interaction*. Teachers provide feedback to students regarding the assessment results, and interventions are planned for students needing additional support.
- *Parent-Teacher Meetings*. Assessment results are shared with parents during parent-teacher meetings where teachers explain performance metrics and suggest improvement strategies and home learning plans.

Chapter II

REVIEW OF EXISTING ALTERNATIVES

The current method used at Pristine Private School demonstrates various shortcomings when tracking student attainment and progress in various assessments. It is time-consuming for large classes, which often involve manual data entry into spreadsheets or similar tools. Teachers manually input each student's scores or raw marks for every internal assessment given in each subject. This will not only take away the valuable time that could be spent on teaching or planning but also slows down the process of analyzing and interpreting data. Another limitation of the current manual approach is its susceptibility to human errors. Mistakes can occur at any stage of the process, from recording scores/raw marks to inputting it in the system. Errors in data could lead to inaccurate tracking of student attainment and progress, and potentially incorrect interventions. Furthermore, identifying and correcting errors is challenging especially when dealing with large volumes of data. Lastly, the current method lacks the ability to perform more sophisticated analyses. For instance, its limited capacity to identify patterns in student performance over time and predict future performance based on trends. This limits the ability of teachers and school administrators to gain a deeper understanding of student performance and to make data-driven decisions.

To cope with these challenges, the school must begin to explore and leverage AI-driven solutions [2][3]. One existing alternative is *School Data Insights*, a web-based subscription service platform that offers data analysis and comprehensive suite of reports to enhance educational strategies [4]. The platform closely aligns with the proposed system, utilizing advanced mathematics and statistics in conjunction with an AI-powered system for data analysis. *School Data Insights* is complemented by professional support from data scientists to enhance the quality and accuracy of the analytical insights provided [4]. Both systems aim to automate data tracking and offer comprehensive analytics, but its pricing model limits the number of users which

is contrary to the school's aim to implement a system that can be easily customized to meet the evolving needs without incurring additional expenses.

Another alternative related system is called *LiveBook* (powered by *Rediker Software*), a comparative analytics technology that helps educators associate content, visualize data, and explore relationships [5]. Pertinent to the proposed system, *LiveBook* uses uploaded assessment data from external sources to create visual charts and tables to see correlations and trends [5]. Additionally, this system focuses on highlighting school's identified strengths, enrollment statistics, and increased achievements, which is in contrast to what the proposed system aims to achieve.

Pupil Progress, a progress tracking and reporting software, closely mirrors the features of the proposed system. It facilitates a deeper understanding of student data by providing live analysis of key subject areas, topics, and skill development [6]. The system utilizes student performance data across subjects, topics and skills to analyze and track progress, make predictions, compare marks or grades between two points in time, and generate relevant reports [6]. However, it lacks a feature to analyze and track student performance based on external benchmark tests, a key focus of the proposed system.

While there are many existing AI-driven solutions and alternative systems for tracking student progress, the proposed system stands out by its customizable nature as it is developed specifically to address the school's needs and offers a more comprehensive approach by uniquely combining automation and leveraging AI to provide more sophisticated data analysis and interpretation [6]. By adopting this innovative solution, the school can ensure data accuracy, improve efficiency, and empower educators with a platform to track student attainment and progress utilizing assessment data from internal and external sources.

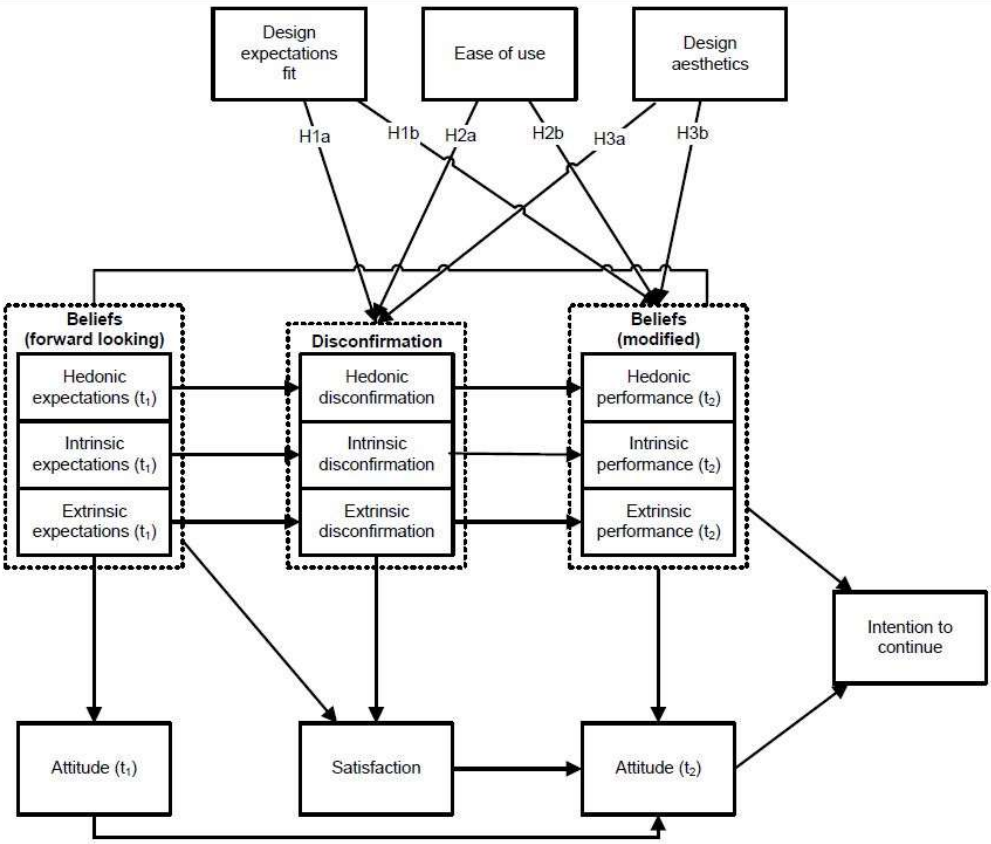
Chapter III

APPROACH TO BE TAKEN IN THIS SUBJECT

A. Theoretical Framework

The Multi-motive Information System Continuance (MISC) model can be effectively utilized in the development of the proposed system. By integrating the key element of this model, the system can be designed to address the various motivational factors that influence the continued use of the system by teachers and school administrators. Research has shown that integrating elements such as perceived usefulness, user satisfaction, and habit formation can significantly impact the sustained engagement of users with information systems [8]. Additionally, intrinsic and extrinsic motivational factors can further enhance user commitment to the system [9][10].

Figure 2. Diagram/schematic of the Multi-motive Information Systems Continuance Model (MISC), from Lowry et al. 2015, p. 525.



In the context of the proposed system, enhance intrinsic motivation key component of MISC could be in the form of interactive dashboards that shows metrics for student performance. Dashboards in the proposed system should be user-friendly and visually appealing to make data analytics more interesting and intuitive. Support extrinsic motivation key component could be in the form of professional development where the developer can offer user trainings and sessions, emphasizing the benefits of using the system. By incorporating enhance intrinsic motivation and supporting extrinsic motivation, the system can be made more engaging and valuable to users. Additionally, providing comprehensive analytics and AI-driven insights can increase perceived usefulness, while ensuring user satisfaction through reliable performance and user feedback loops, such as collecting and incorporating user feedback on a regular basis to continually improve the system, ensuring it meets users' needs and expectations. This approach ensures that the proposed system maintains high-level use satisfaction and adoption, ultimately contributing to its long-term success.

Developing the proposed system to be robust involves integrating several key design principles. These design principles can be incorporated into the proposed system through the following approaches:

- **Modular Design**. Developing the proposed system will adopt the modular approach where the system is divided into modules, such as user management, data import facility, analytics, and reporting.
- **User-Centered Design**. Engage teachers and school administrators during the design phase to ensure the system meets their needs and enhances usability.
- **Scalability**. The system will be designed to scale horizontally to handle an increasing number of system users and vertically to handle the increasing volume of data, expecting increasing cohort enrollees annually.
- **Agile Methodology**. Develop the system to use Agile methodology to ensure continuous improvement and adaptation based on user feedback.

B. Rationale for the Framework

Tracking attainment and progress in Year 2 assessments, both internal and external benchmarks, is currently accomplished using the conventional method of formula-based calculations and spreadsheets. The long-term use of this approach presents several significant challenges that may hinder the school's ability to effectively monitor and support student performance over time.

The Multi-motive Information System Continuance (MISC) model is best suited and adopted to help improve the current conventional method by aligning the design and features of the proposed system with key intrinsic and extrinsic motivational factors, making it not just highly functional but also intuitive and engaging for its users. MISC model is a well-established theoretical framework that provides a comprehensive approach to understanding and enhancing the long-term use of information systems [11].

Adopting the Multi-motive Information System Continuance (MISC) model contributes to making the application exceptionally user-friendly, aligning the system's functionalities and user interactions with user expectations ensures a straightforward and accessible experience for all users.

C. Technologies Used

Several technologies will play crucial roles in different aspects of system development. The backend will be built using Flask, a lightweight Python web framework [12]. Flask-RESTful, an extension for Flask, will be used to create RESTful API endpoints that handle CRUD (Create, Read, Update, Delete) operations [13]. These endpoints will facilitate seamless communication between the backend and the frontend. Additionally, Flask's built-in Jinja2 templating engine will be employed for server-side template rendering, delivering HTML pages that include data directly from the backend [14]. Python libraries such as Pandas, NumPy, and SciPy will be leveraged for data manipulation and statistical analysis.

Open AI's ChatGPT will be integrated into a Flask-RESTful API through a Flask route that sends input (assessment data) in the form of prompts, passing it over to the OpenAI's Embeddings API, and returning the generated response from OpenAI's ChatCompletions API (GPT analysis and interpretation) back to the system [15]. Python library such as `openai` library will provide an interface to OpenAI's APIs (Embeddings and ChatCompletions) and incorporate the API call in Flask-RESTful API's logic [16].

On the front end, web technologies such as HTML5, CSS3, and JavaScript will be used to build the user interface. Front end web framework like Vue.js will be utilized to create interactive user interfaces and dashboards that will include dynamic charts, graphs, and tables [17]. Additionally, libraries like Plotly.js or D3.js will be used for generating customizable and visually appealing visualizations [18].

Flask extensions like Flask-Login or Flask-Security will be utilized for user authentication, authorization, and session management. Flask libraries like ReportLab [19] will also be used to provide users with options to download reports in various formats (e.g. PDF, CSV).

Database management is facilitated through SQLAlchemy, a widely used Object-Relational Mapping (ORM) tool in Python. Through SQLAlchemy's intuitive API, CRUD operations can be performed on the database [20].

To ensure consistency and ease of deployment during the different stages of the development cycle, containerization will be applied and will use Docker as the primary containerization tool. Deployment will be accomplished through a continuous integration/continuous deployment (CI/CD) pipeline which will be set up using GitHub Actions. Version control for the codebase will be handled using a version control system like Git will be utilized. The codebase will be hosted in a Git repository like GitHub.

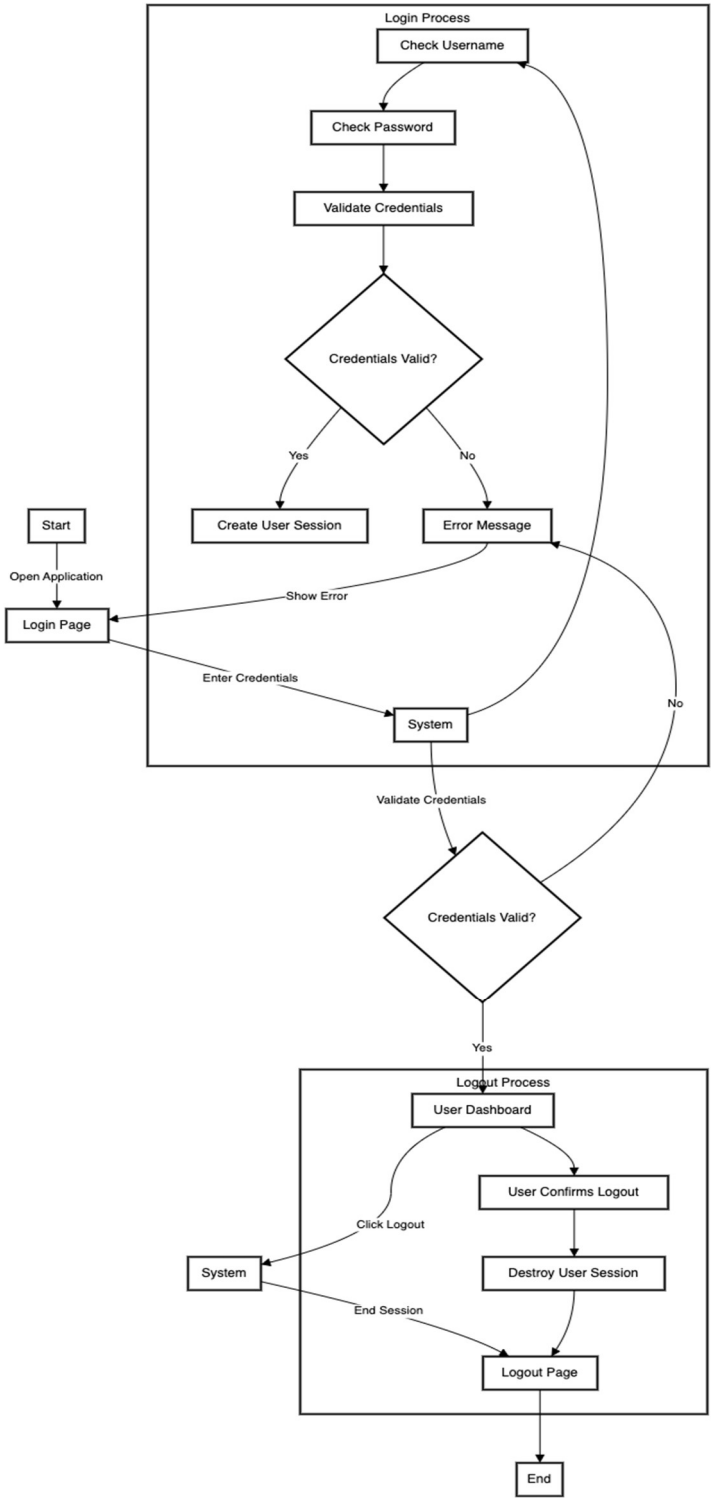
Cloud hosting will be provided for the proposed system through Amazon Web Services (AWS) Elastic Cloud Compute (EC2) [21]. Docker container will be deployed in Amazon ECS (Elastic Container Service). Amazon S3 (Simple Storage Service) will be utilized to provide scalable object storage solution for storing static assets, such as images, documents, multimedia files associated with the proposed system [22]. AWS CloudWatch will be integrated to provide monitoring on performance and health of the application and logging services for troubleshooting and optimization purposes [23].

Chapter IV

PROJECT PLAN

A. Concept

Figure 3. Process Flow Diagram for Authentication and Session Management



The web application requires users (teachers/school administrators) to provide valid credentials, such as username and password, when logging in to the system. System checks are in place for security. If credentials are invalid, error message is displayed, and the user remains in the login page. Otherwise, the user will have access to the system and access the dashboard right after login.

Upon clicking the logout button, the system ends the user's session, and the user is directed to the logout confirmation page.

Figure 4. Process Flow Diagram for Roles and Access Control

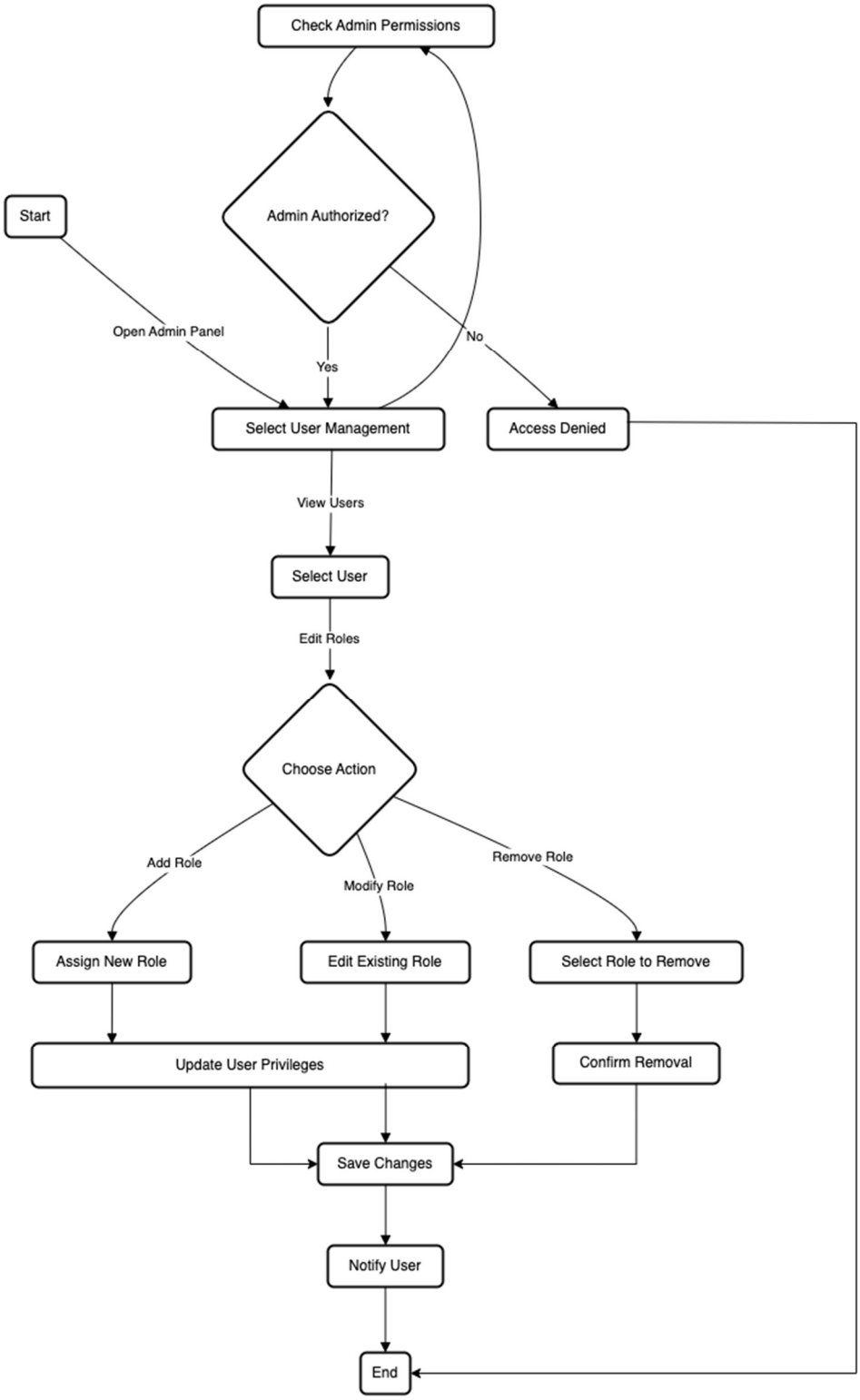


Figure 4 outlines the steps a system admin takes to add, modify, or remove user roles, ensuring that the right permissions are in place and users are informed of changes.

Figure 5. Process Flow Diagram for User Control

The user control process flow involves verifying that correct credentials are in place before granting access to system services intended for authorized users.

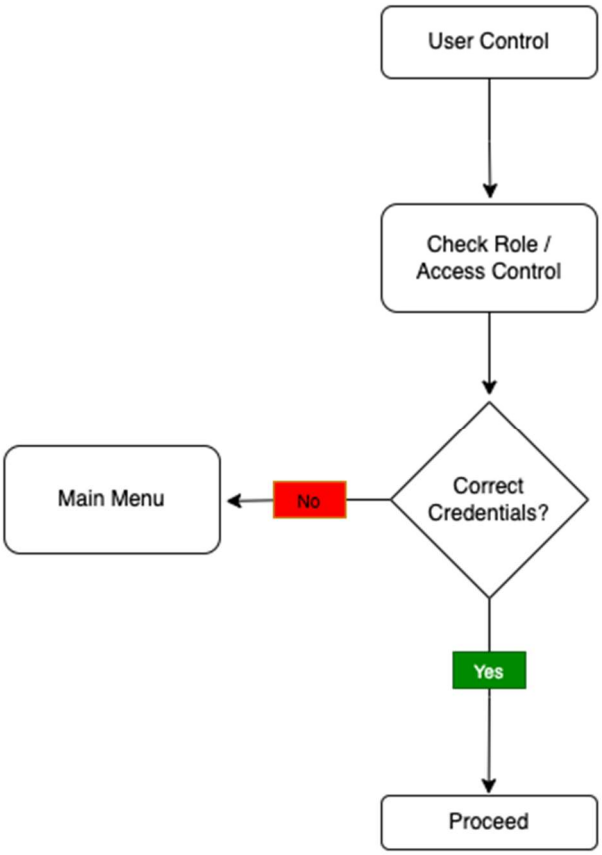
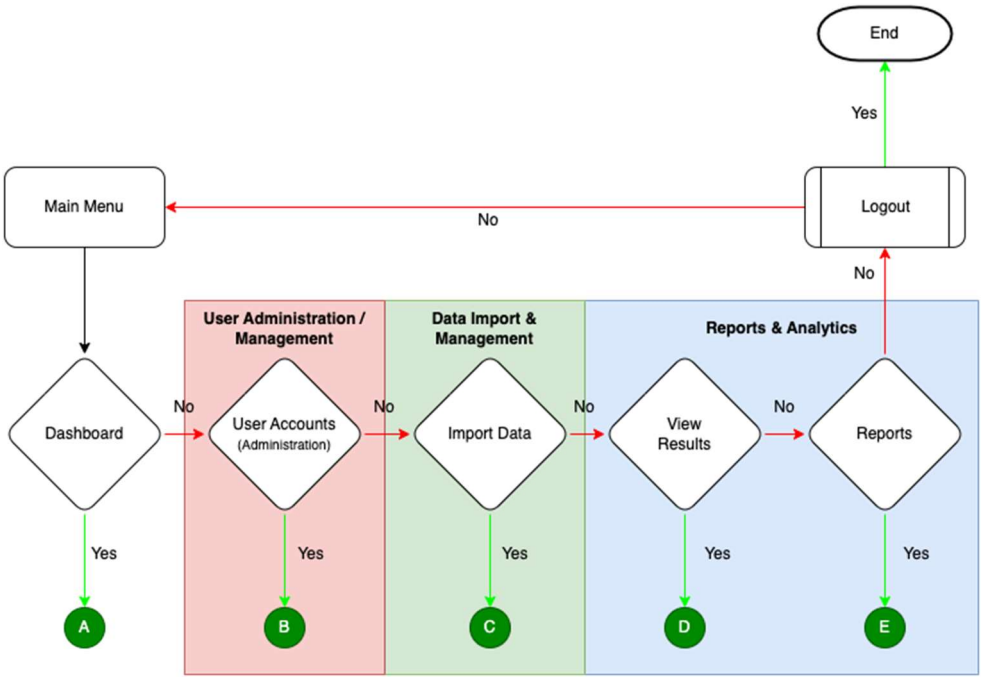
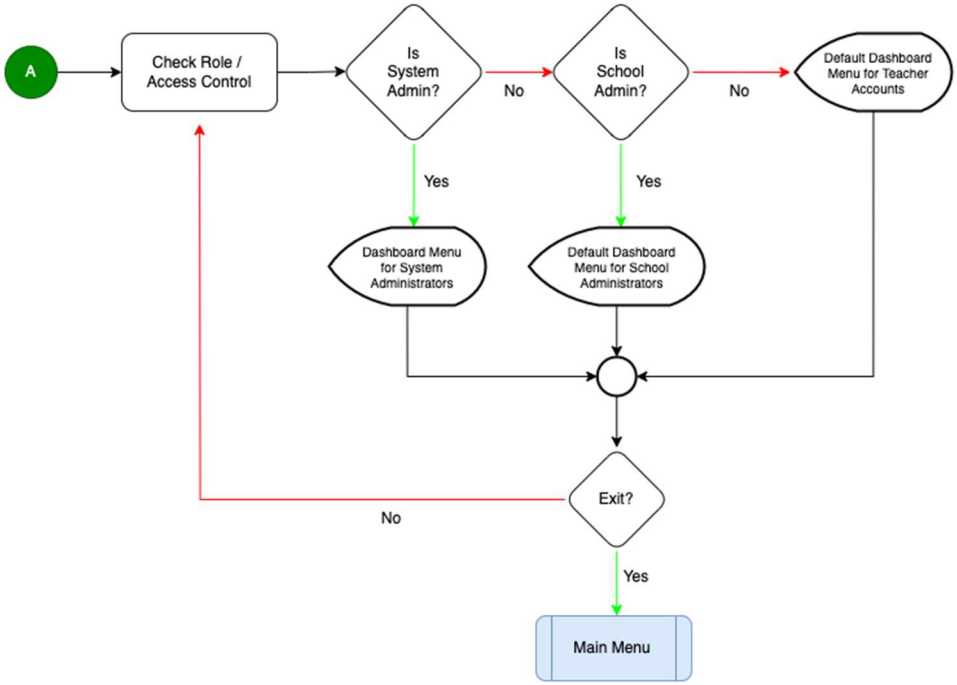


Figure 6. Process Flow Diagram for Main Menu



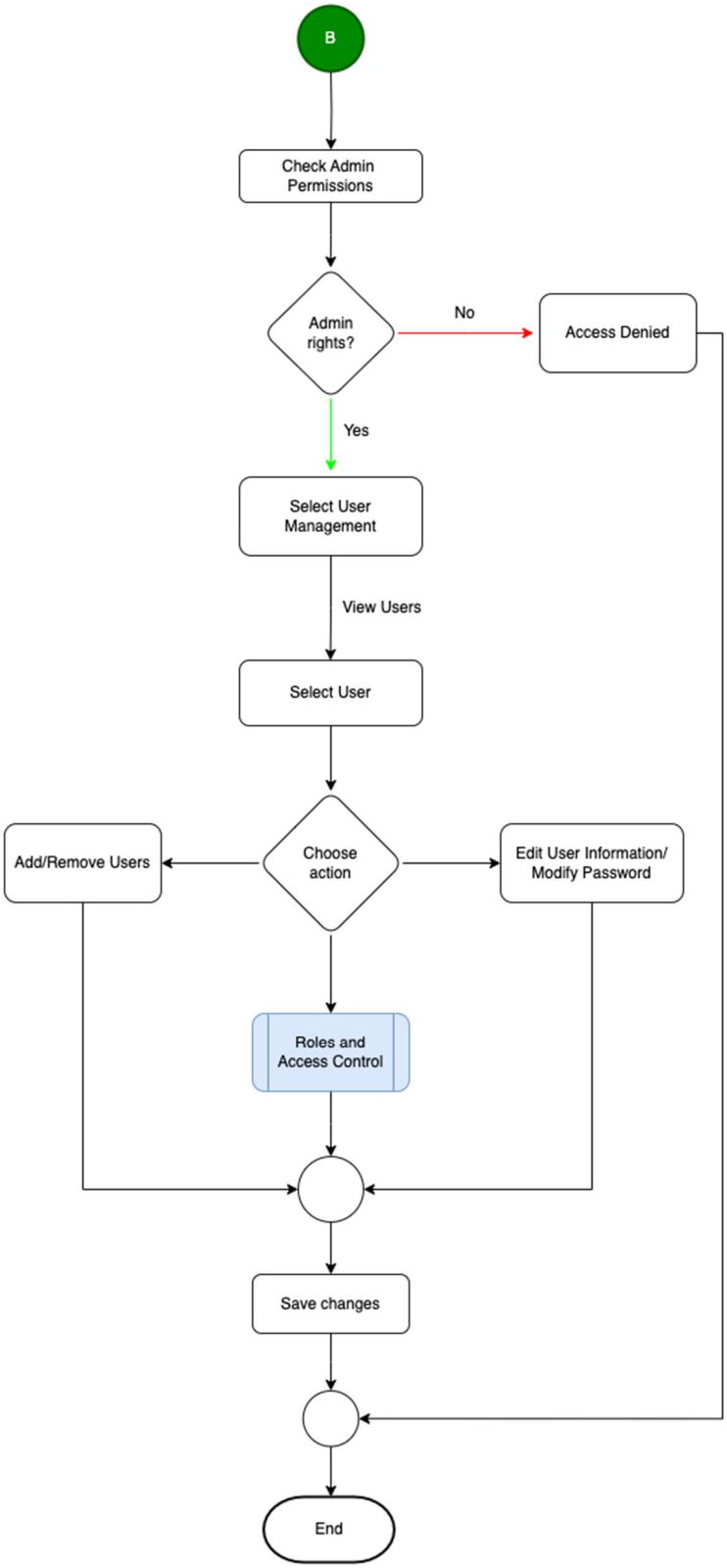
The main menu of the system involves six (6) different options. These menu choices contain sub-menus which present options according to user role and privileges.

Figure 7. Process Flow Diagram for Dashboard Facility



Dashboard facility displays menu according to user role/access control. There are three (3) user types/roles within the system: system admin, teacher, and school administrators. Each one has different menu options within the dashboard.

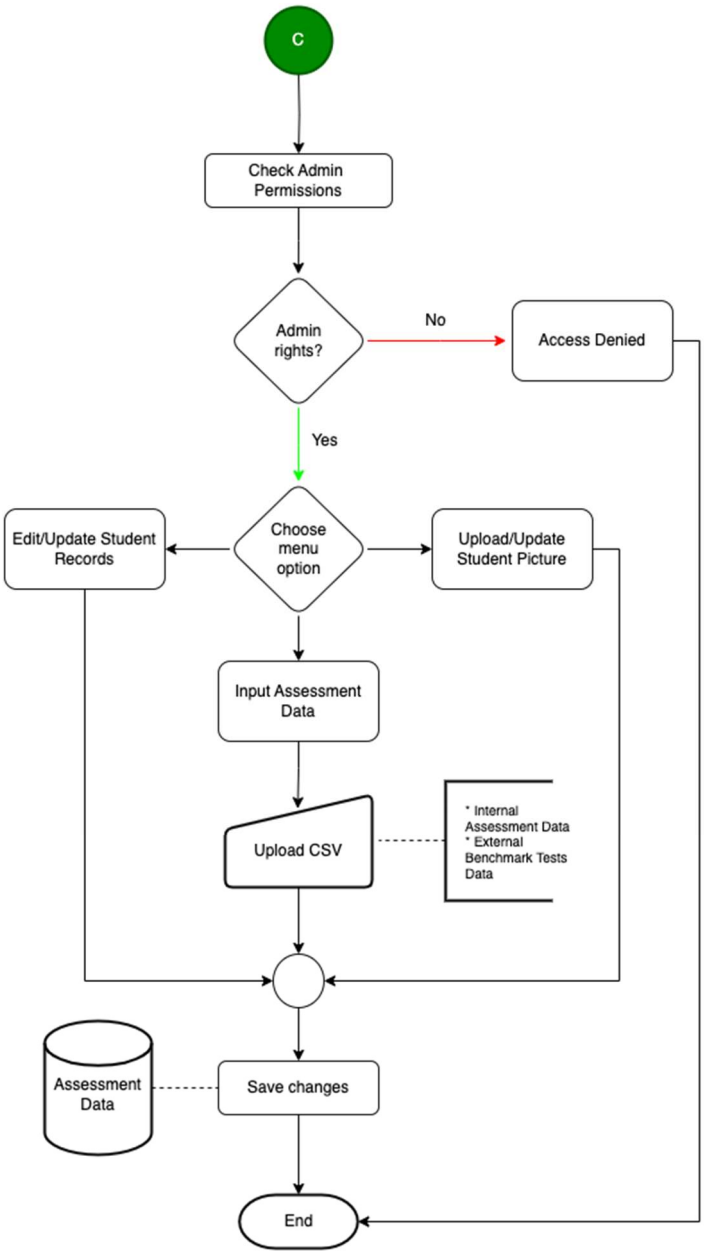
Figure 8. Process Flow Diagram for User Administration/Management



The user administration and management module contain essential components that deals with user data and access. Only users with admin rights

can access the module. Within the module, system admin can insert/delete users, modify roles/access controls of existing users, and edit user information and modify password.

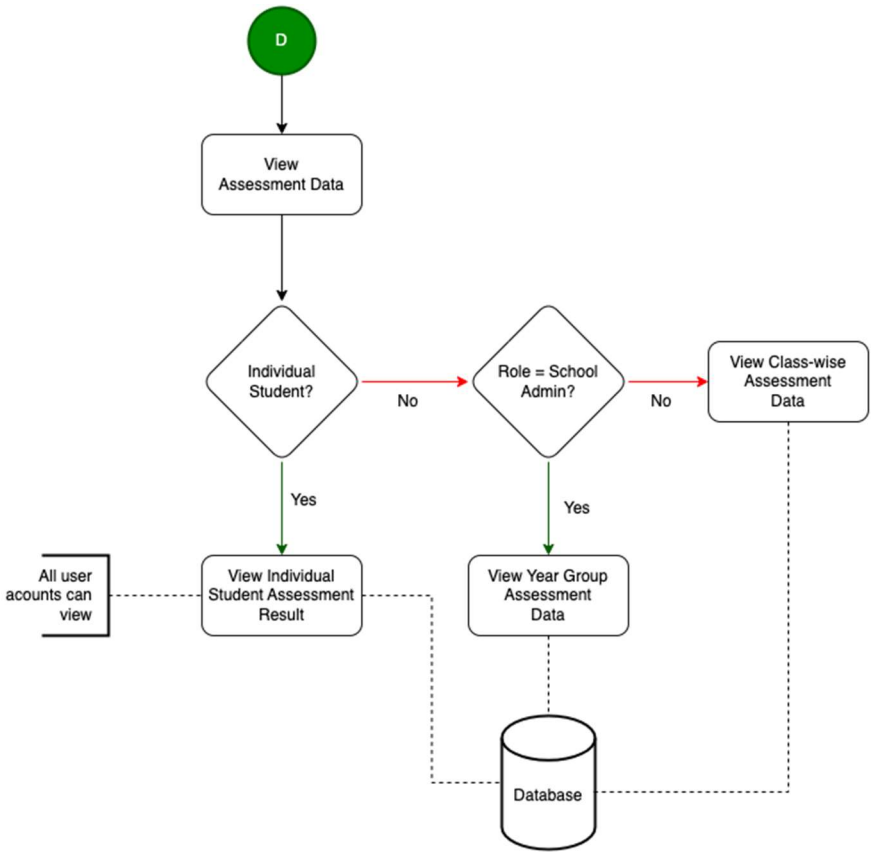
Figure 9. Process Flow Diagram for Data Integration and Management



The process flow diagram for data integration and management begins with checking admin privileges to ensure that only authorized users will have access to services within the module. Furthermore, it includes uploading CSV/Excel file containing assessment data, both internal and external benchmark tests, along with some information about the students. This allows for the bulk addition of student records, providing streamlined method to initialize the database with

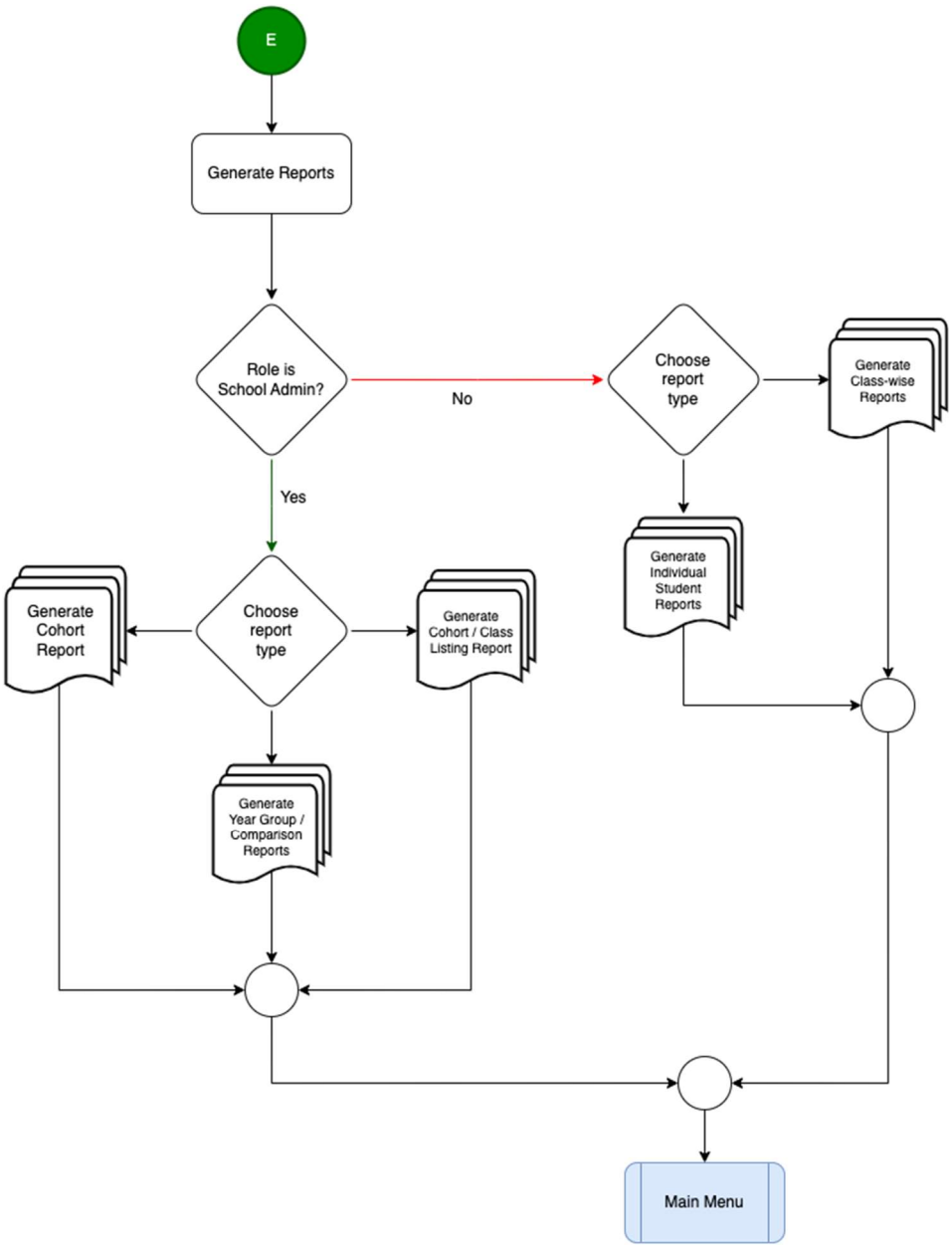
new information. The diagram also reflects modifying existing entries uploading/editing student picture and editing/updating student records to ensure they are current and accurate information.

Figure 10. Process Flow Diagram for Reports and Analytics – View Results



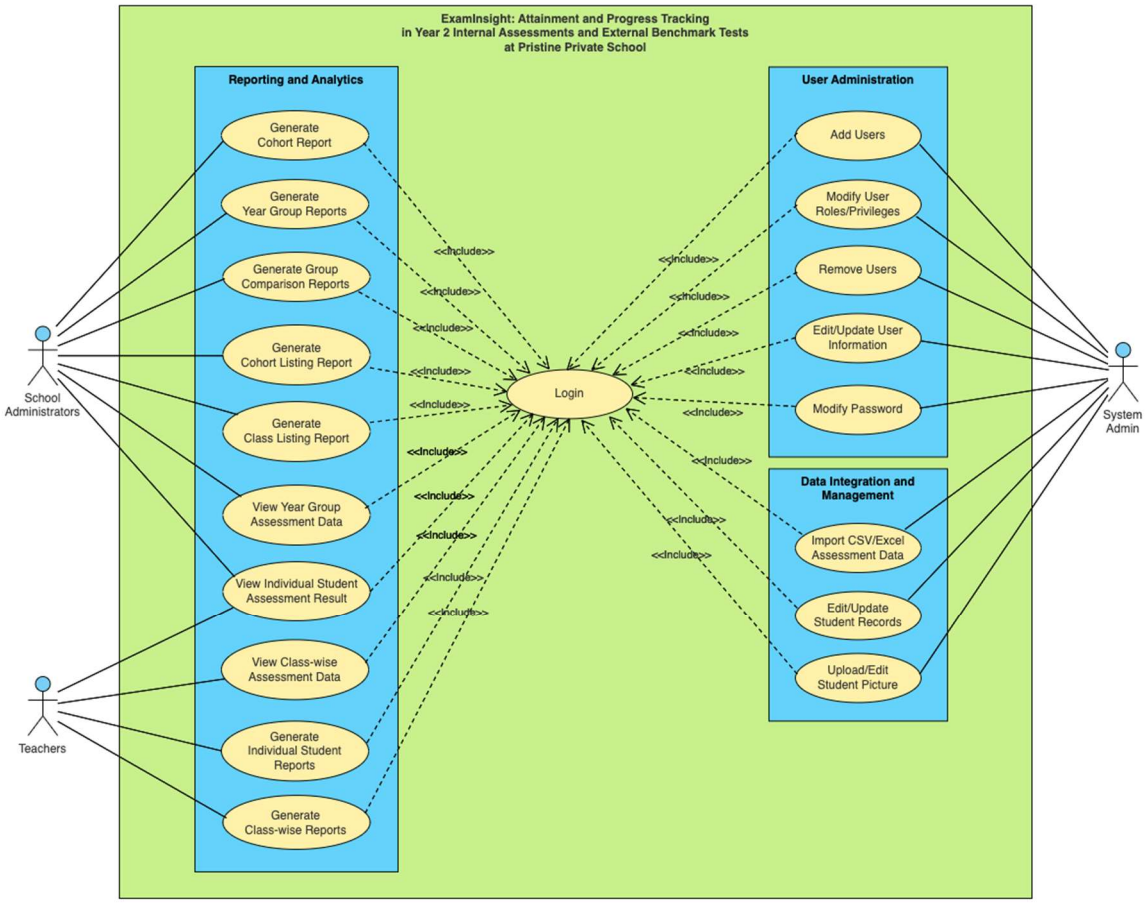
The figure illustrates that individual student assessment data can be viewed by any user account, without requiring system or school admin roles. However, accessing year group and class-wise assessment results needs the correct access role and user privileges.

Figure 11. Process Flow Diagram for Reports and Analytics – Generate Reports



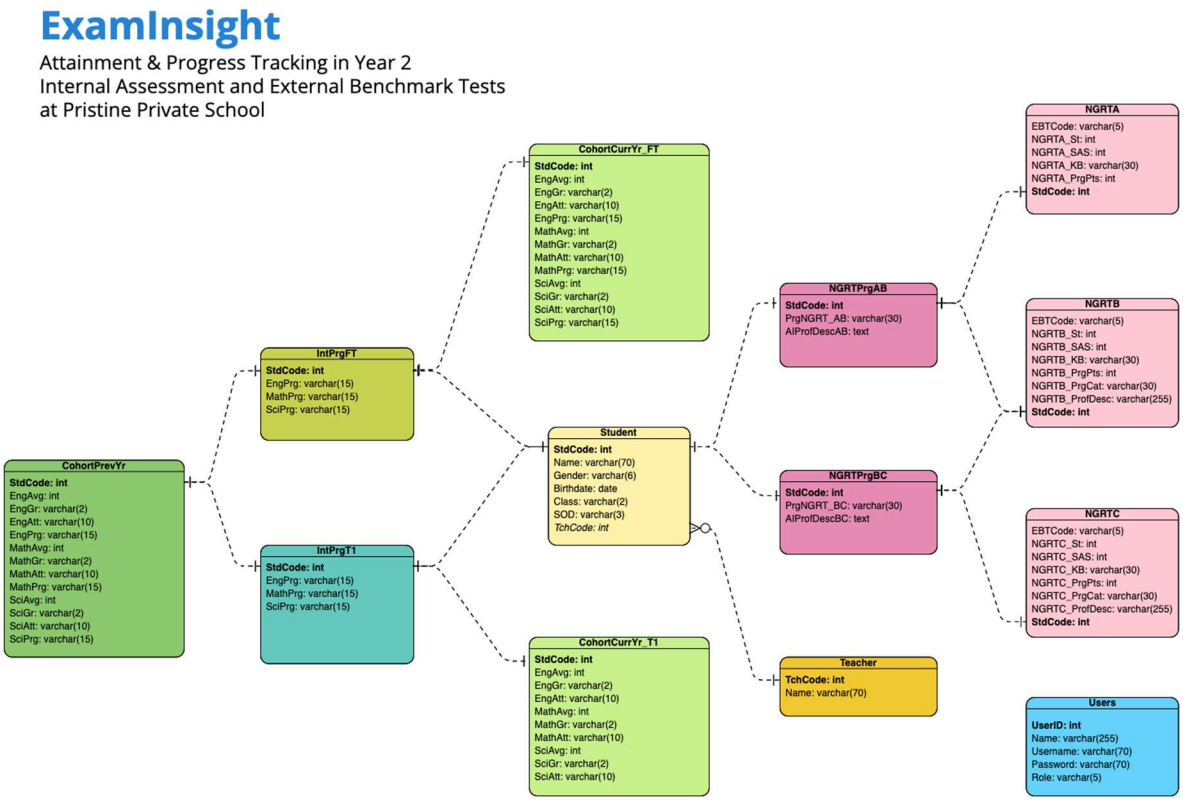
The diagram illustrates that generating reports within the system requires the appropriate user role or privileges. Furthermore, it highlights that school administrators have access to a more comprehensive set of reports compared to regular users, such as teachers

Figure 12. Use Case Diagram of the Proposed System



This Use Case diagram provides insights on the intricate architecture the proposed system. The interconnected arrows emphasize how different functionalities rely on each other. Additionally, the diagram shows that different end-user roles/privileges will have different access to system functionalities.

Figure 13. Entity-Relationship Diagram of the System



This entity-relationship diagram (ERD) represents the fundamental database structure for the proposed system. It visually outlines how data will be organized and stored within the database. The diagram is divided into two main sections: one for internal assessments (highlighted by green and blue boxes) and another for external benchmark tests (indicated by pink and yellow boxes).

B. Methods

An agile project management technique, specifically employing personal SCRUM practices within a phrased approach will be followed to develop the proposed system. System users and other stakeholders will be engaged during the development of the proposed system though constant user feedback loop and interviews. From this data gathering, the developer will point out the use of AI technology to enhance the current conventional method of tracking attainment and progress in year 2 assessments, both internal and external benchmark.

C. Plan for User Testing and Project Assessment

The plan for user testing and project assessment for the proposed system will be meticulously structured to ensure thorough evaluation and optimal functionality. Additionally, a diverse group of end-users (teachers and school administrators) will be engaged in a series of user acceptance tests (UAT) to provide valuable feedback on usability, performance, and overall user experience. Both functional and non-functional testing will be conducted to ensure all features work as intended and to assess aspects such as load times and system stability. Post-testing, the feedback collected will be systematically analyzed, and necessary adjustments will be made to address any identified issues or enhancements.

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